

PrepPilot — Technology Stack

Senior project · Next.js 16 · FastAPI · 8 LLM models · Sprint 5.7 (2026-05-18)

Frontend



Next.js 16
App Router



React 19
Server + Client



TypeScript 5
strict mode



Tailwind v4
+ typography



Prisma 6
+ NextAuth v5

AI / LLM



OpenAI
GPT-5.4 nano ★
+5 / +mini / 4o

Anthropic
Claude

Opus 4.7 · Sonnet 4.6 · Haiku 4.5



LangChain
cached factory
+history truncate



MongoDB
prod · Atlas



SQLite
dev · file



Prisma ORM
multi-provider

Backend



FastAPI
async routes
Pydantic v2



Python 3.13
uvicorn



scikit-learn
27 algorithms



NumPy
vectorized ops



LightGBM
+ XGBoost

Hosting



Vercel
frontend
edge functions



Render
backend primary
render.yaml



Fly.io
backend alt
fly.toml



Docker
+ Compose
local + prod



Region
closest to TH



GitHub
4 repos